

Show understanding of the purpose and benefits of networking devices

9608_s21_qp_11_sc_9.b

QUESTION

- (b) Complete the table by identifying the **most appropriate** term for each description. Each term must be different.

Description	Term
Receives data packets from a network and forwards them onto a similar network	
Manages access to a centralised resource	
Joins networks that use different sets of rules to transmit data	
Monitors and controls incoming and outgoing network traffic based on set criteria	

[4]

MARK SCHEME

9(b)	1 mark for each correct term	4										
<table><tr><th>Description</th><th>Term</th></tr><tr><td>Receives data packets from a network and forwards them onto a similar network</td><td>Router</td></tr><tr><td>Manages access to a centralised resource</td><td>Server</td></tr><tr><td>Joins networks that use different sets of rules to transmit data</td><td>Gateway</td></tr><tr><td>Monitors and controls incoming and outgoing network traffic based on set criteria</td><td>Firewall</td></tr></table>			Description	Term	Receives data packets from a network and forwards them onto a similar network	Router	Manages access to a centralised resource	Server	Joins networks that use different sets of rules to transmit data	Gateway	Monitors and controls incoming and outgoing network traffic based on set criteria	Firewall
Description	Term											
Receives data packets from a network and forwards them onto a similar network	Router											
Manages access to a centralised resource	Server											
Joins networks that use different sets of rules to transmit data	Gateway											
Monitors and controls incoming and outgoing network traffic based on set criteria	Firewall											

QUESTION

- (c) When Bart is at work, he connects his work laptop to his employer's Local Area Network (LAN). The LAN has both a router and a gateway.

Give **two** similarities and **one** difference between a router and a gateway.

Similarity 1

.....

Similarity 2

.....

Difference

.....

.....

[3]

MARK SCHEME

8(c)	1 mark for each similarity, max 2 - Both devices regulate network traffic between two networks // connect two networks - Both receive packets from a network and both forward packets onto a network 1 mark for a difference - A Router connects two networks using the same protocol, a Gateway can connect two networks using different protocols	3
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QUESTION

(c) A network can use routers and gateways.

Explain the role of routers **and** gateways in a network.

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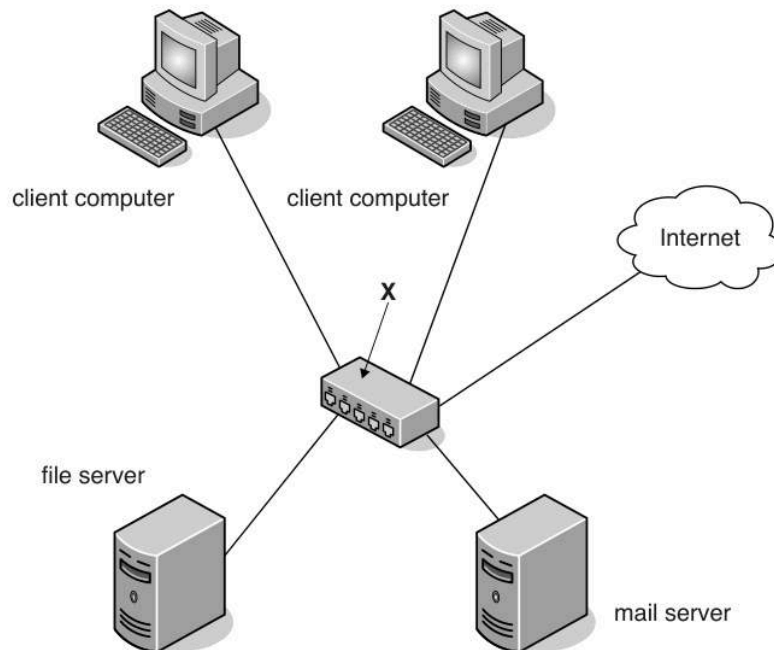
..... [4]

MARK SCHEME

1(c)	1 mark per bullet point to max 4, max 3 for router, max 3 for gateway Only award the repeated bullet points (1 to 5 in each section) once Router: • Connects two (or more) networks • Can connect a network to a WAN // acts as the single access point for... • Receives packets and forwards towards the destination • ...using the IP address of the destination • Assigns private IP addresses • Operates between <u>similar</u> networks // networks using the <u>same protocol</u> • Can be used to segment a network Gateway: • Connect two (or more) networks • Can connect a network to a WAN // acts as the single access point for... • Receives packets and send packets towards the destination • ...using the IP address of the destination • Assigns private IP addresses • Connects two <u>dissimilar</u> networks // networks that use <u>different protocols</u>	4
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QUESTION

(b) The diagram shows a computer network with connection to the Internet.



Name the hardware device labelled X.

.....[1]

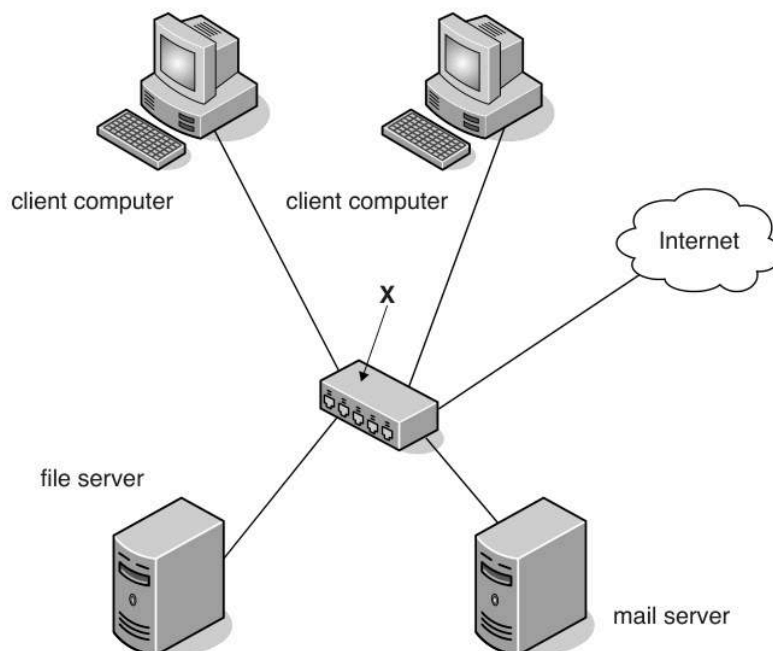
MARK SCHEME

(b) router

[1]

QUESTION

(b) The diagram shows a computer network with connection to the Internet.



Name the hardware device labelled X.

.....[1]

MARK SCHEME

(b) router

[1]

QUESTION

(c) Name the hardware device that is being described:

- (i) A device that transfers data from one network to another in an intelligent way. It has the task of forwarding data packets to their destination by the most efficient route.

.....[1]

MARK SCHEME

(c) (i) router

[1]

9608_s15_qp_12_sc_5.c.ii

QUESTION

- (ii) A device used between two dissimilar LANs. The device is required to convert data packets from one protocol to another.

.....[1]

MARK SCHEME

(ii) gateway

[1]

9608_s15_qp_11_sc_5.c.ii

QUESTION

- (ii) A device used between two dissimilar LANs. The device is required to convert data packets from one protocol to another.

.....[1]

MARK SCHEME

(ii) gateway

[1]

Show understanding of the characteristics of a LAN (local area network) and a WAN (wide area network)

Explain the client-server and peer-to-peer models of networked computers

9608_w21_qp_11_sc_3

QUESTION

- 3 Server-side scripting is used to access a company database over the Internet.

Steps 1 to 6 describe the sequence of events.

Four of the statements **A**, **B**, **C**, **D**, **E** and **F** are used to complete the sequence.

Letter	Statement
A	The web server processes the PHP code.
B	The server renders the HTML code and displays the web page.
C	The browser renders the HTML code and displays the web page.
D	The web server returns the (HTML) web page to the client web browser.
E	The web server accesses the page from its hard disk.
F	The web browser processes the PHP code.

Write **one** of the letters **A** to **F** in each appropriate row to complete the sequence.

1. The browser requests the web page from the web server.
2.
3.
4. The web server produces the HTML code for the web page.
5.
6.

[4]

MARK SCHEME

3	One mark for each correct letter in the correct position	4
	1 The browser requests the web page from the web server.	
	2 E	
	3 A	
	4 The web server produces the HTML code for the web page.	
	5 D	
	6 C	

QUESTION

11 The Internet uses a client-server model.

(a) Describe the role of clients and servers on the Internet.

.....

.....

.....

..... [2]

MARK SCHEME

11(a)	One mark per bullet point to max 2 • Web pages/files are saved on servers • Clients send requests to the web servers • Web servers process the requests • ...and return the results to the client • ... client displays the results to the user	2
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QUESTION

6 (a) A customer completes a booking form on a web page and clicks a submit button to submit the form.

The following sequence (1 to 6) describes the steps that take place when the form is submitted.

- 1
- 2 The form data is transmitted to the web server
- 3
- 4
- 5 The HTML code is returned to the browser
- 6

Write **one** of the letters A to D in each row (1, 3, 4 and 6) to complete the sequence.

A	The browser displays the web page
B	Server-side code is processed
C	Client-side code is processed
D	The web server produces the HTML code

[3]

MARK SCHEME

6(a)	1 mark per bullet point • C in correct position • B and D in correct positions • A in correct position 1 C // The client-side code is processed 2 The form data is transmitted to the web server 3 B // The server-side code is processed 4 D // The web server produces the HTML code 5 The HTML code is returned to the browser 6 A // The browser displays the web page	3
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9608_s21_qp_12_sc_5.c

QUESTION

(c) The page is loaded and the values 2 and 8 are entered.

State the output when the calculate button is clicked.

.....

..... [1]

MARK SCHEME

5(c)	The area is: 8	1
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9608_s21_qp_12_sc_5.d

QUESTION

(d) State the meaning of the || operator in line 15 of the code.

..... [1]

MARK SCHEME

5(d)	(logical) OR	1
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QUESTION

Cambridge International Holidays allows customers to make holiday bookings on its website.

(f) Bochen visits the Cambridge International Holidays website to book a holiday.

The sequence (1 to 8) below describes the steps that take place when he submits a booking.

Four of the statements **A**, **B**, **C**, **D**, **E** and **F** are used to complete the sequence.

A	Any errors found at the server side are flagged, and step 1 is repeated.
B	HTML code is used on the client's web browser to validate the form data.
C	PHP code is executed to generate a confirmation (HTML) web page that is returned to the client's web browser.
D	The form data is transmitted to Cambridge International Holidays' web server.
E	JavaScript code is executed to generate a confirmation (HTML) web page that is returned to the client's web browser.
F	JavaScript code is executed on the client's web browser to validate the form data.

Write **one** of the letters (**A**, **B**, **C**, **D**, **E** or **F**) in the appropriate row to complete the sequence.

- 1 Bochen completes the online booking and clicks 'Submit'.
- 2
- 3 Any errors found are flagged, and step 1 is repeated.
- 4
- 5 PHP code is executed to perform extra data validation checks on the form data.
- 6
- 7 The booking details are added to the database.
- 8

[4]

MARK SCHEME

5(f)	1 mark for each correct letter/statement in each space 1 Bochen completes the online booking form and clicks 'Submit'. 2 F // JavaScript is executed on the client's web browser to validate the form data. 3 Any errors found are flagged, and step 1 is repeated. 4 D // The form data is transmitted to Cambridge International Holidays' webserver. 5 PHP code is executed to perform extra data validation checks on the form data. 6 A // Any errors found at the server side are flagged and step 1 is repeated. 7 The booking details are added to the database. 8 C // PHP is executed to generate a confirmation (HTML) web page that is returned to the client's web browser.	4
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QUESTION

6 A web page includes HTML, JavaScript and PHP code.

```

01 <html>
02 <body>
03
04 <p>
05 <?PHP
06     $message = "Enter the current time"
07     echo $message
08 ?>
09 </p>
10
11 24 hour clock hour<input id = "Time" value = "">
12 <button onclick = "timeOfDay()">Enter</button>
13
14 <script>
15     function timeOfDay() {
16         var hour, greeting;
17         hour = document.getElementById("Time").value;
18         if (hour < 18) {
19             greeting = "Good day";
20         } else {
21             greeting = "Good evening";
22         }
23         alert(greeting);
24     }
25 </script>
26
27 </body>
28 </html>

```

(a) The page is loaded and the value 16 is entered.

State the output when the enter button is clicked.

..... [1]

MARK SCHEME

6(a)	Good day	1
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9608_s21_qp_11_sc_6.b

QUESTION

(b) Give the line number where the JavaScript function is called.

..... [1]

MARK SCHEME

6(b)	12	1
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9608_s21_qp_11_sc_6.c

QUESTION

(c) Give the identifier names of **two** variables used in the JavaScript code.

1
2 [1]

MARK SCHEME

6(c)	1 mark for both answers hour greeting	1
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9608_s21_qp_11_sc_6.d

QUESTION

(d) Tick (✓) **one** box in each row to indicate whether each of the following statements is true or false.

Statement	True	False
The program contains client-side and server-side code		
The PHP code in the program will run on the client-side		
Line 19 of the code outputs the message "Good day"		
Line 18 of the code contains a conditional statement		

[2]

MARK SCHEME

6(d)	1 mark for each pair of correct answers (shaded): <table border="1"> <thead> <tr> <th>Statement</th><th>True</th><th>False</th></tr> </thead> <tbody> <tr> <td>The program contains client-side and server-side code</td><td>✓</td><td></td></tr> <tr> <td>The PHP code in the program will run on the client-side</td><td></td><td>✓</td></tr> <tr> <td>Line 19 of the code outputs the message "Good day"</td><td></td><td>✓</td></tr> <tr> <td>Line 18 of the code contains a conditional statement</td><td>✓</td><td></td></tr> </tbody> </table>	Statement	True	False	The program contains client-side and server-side code	✓		The PHP code in the program will run on the client-side		✓	Line 19 of the code outputs the message "Good day"		✓	Line 18 of the code contains a conditional statement	✓		2
Statement	True	False															
The program contains client-side and server-side code	✓																
The PHP code in the program will run on the client-side		✓															
Line 19 of the code outputs the message "Good day"		✓															
Line 18 of the code contains a conditional statement	✓																

9608_s21_qp_11_sc_6.e

QUESTION

(e) Examine the following list of terms:

bandwidth	browser	compiler	HTML	interpreter
JavaScript	PHP	router	server	server-side
transmission	validation			

Complete the following sentences by filling in the missing terms from the list.
Some terms may be used more than once. Some terms are not used.

..... can be performed both client-side and server-side.

It is performed more rapidly by the because there is no delay in
transmitting and receiving data to and from the

It is also performed on the server-side, because the client's may
not support, so the data will still need checking to avoid errors.

[5]

MARK SCHEME

6(e)	1 mark for each correctly completed term Validation can be performed both client-side and server-side. It is performed more rapidly by the browser because there is no delay in transmitting and receiving data, to and from the server. It is also performed server-side because the client's browser may not support JavaScript, so the data will still need checking to avoid errors.	5
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QUESTION

- (b) Explain how the client-server model enables the employees to access the same files from different computers.

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..... [2]

MARK SCHEME

1(b)	1 mark per bullet point to max 2 • The employees' computers are the clients • The server hosts the (shared) files • An employee can request a file (from the server) from any of the client computers • Several employees can access the same file (on the server) at the same time	2
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QUESTION

(c) A customer enters a song title into a web page to listen to the song. The design of the web page is shown:

Company Name	
Navigation Bar	
Enter song title:	
	Search

The web page will make use of both client-side and server-side scripting.

(i) Explain how client-side scripting will be used in this web page.

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.....
.....
..... [3]

MARK SCHEME

7(c)(i)	1 mark per bullet point to max 3 • The web page may have interactive features • Created using JavaScript • E.g. text box, buttons • Validates the input client-side • Handles the data returned from the server-side script	3
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QUESTION

- (ii) Explain how server-side scripting will be used after the customer clicks the 'Search' button.

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..... [2]

MARK SCHEME

7(c)(ii)	1 mark per bullet point to max 3 • Database on the server is accessed • Using PHP • Searches for the data the user entered • Returns the song // Returns a message to say song not found	2
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QUESTION

- (c) The self-checkout machines connect to a server that stores all the data for the supermarket. This is a client-server network.

- (i) Describe, using an example for the supermarket, the client-server network model.

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..... [4]

MARK SCHEME

1(c)(i)	1 mark per bullet point to max 3 plus 1 mark for suitable example When a barcode on an item is scanned ... the server performs any requested tasks // the server looks up the details of the product The self-checkout machine is a client ... that send requests to the server // the self-checkout machine asks for, e.g. the price of the item The server returns the results of the request // the server returns e.g. the item price Self-checkout machine displays e.g. price to the user	4
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9608_w19_qp_11_sc_4.a.i

QUESTION

4 Customers of a bank can access their account information by logging in on the bank's website.

(a) The bank has a client-server model of networked computers.

(i) Describe, using the bank as an example, the key features of a client-server model.

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..... [3]

MARK SCHEME

4(a)(i)	1 mark per bullet point to max 3. If no application to the bank max 2 The bank's server holds the customer account data / website ... and performs the requested tasks / processes. The computers used by the customers are the clients that send requests to the server ... which returns the results of the request ... E.g. a customer asks for a list of recent transactions on their account.	3
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9608_w19_qp_11_sc_4.a.ii

QUESTION

(ii) Give **two** other examples of applications that can use the client-server model.

- 1
-
- 2
-
- [2]

MARK SCHEME

4(a)(ii)	1 mark per example to max 2 e.g. Sending and receiving email A company or school centrally storing files Using a print server Using a file server	2
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9608_w19_qp_11_sc_4.b

QUESTION

(b) The bank's customers log in to the website using a web application.

Explain why the web application uses server-side scripting.

-
-
-
-
-
-
- [3]

MARK SCHEME

4(b)	1 mark per bullet point to max 3 All data is held on the server // All processing is performed on the server The server only sends the results of the query to the client The client does not have access to all the data ... which keeps the data more secure / consistent Customers can be identified when they log in ...from a database of usernames and passwords	3
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9608_s19_qp_12_sc_1.d

QUESTION

(d) The company has an email server.

Identify **three** other types of server.

1

2

3 [3]

MARK SCHEME

1(d)	1 mark per bullet point to max 3 for any valid answer For example: • File server • Print server • Proxy server • Web server • Application server	3
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9608_w18_qp_11_sc_6.c

QUESTION

(c) Explain the purpose of the code in line 18.

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..... [2]

MARK SCHEME

6(c)	1 mark per bullet point Passes the value in \$message / message variable back to the code that called it // ... to replace the function call	2
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9608_w18_qp_11_sc_6.d

QUESTION

(d) The PHP code in a web page uses server-side scripting.

List the sequence of events that take place when a user requests a web page containing PHP code.

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.....[4]

MARK SCHEME

6(d)	1 mark per bullet point to max 4 The (client web) <u>browser</u> requests the web page from the web server The web server accesses the page (from its hard drive) The web server processes / executes the PHP code and produces the HTML for the web page // ... the web server creates the web page The web server returns the (HTML) web page to the client web <u>browser</u>	4
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9608_s18_qp_13_sc_1.d

QUESTION

(d) State whether this PHP code will run client-side or server-side.

.....[1]

MARK SCHEME

1(d)	Server-side	1
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9608_s18_qp_12_sc_6.a

QUESTION

6 A web page includes the following HTML and JavaScript code.

```

01  <html>
02  <body>
03
04  <p>Enter your mark</p>
05  <input id="Mark" value="0">
06  <button onclick="calcGrade()">Enter</button>
07
08  <script>
09      function calcGrade() {
10          var mark, grade;
11          mark = document.getElementById("Mark").value;
12          if (mark >= 90) {
13              grade = "A"
14          } else if (mark >= 80) {
15              grade = "B"
16          } else if (mark >= 70) {
17              grade = "C"
18          } else if (mark >= 60) {
19              grade = "D"
20          } else if (mark >= 50) {
21              grade = "E"
22          } else {
23              grade = "U"
24          }
25          alert("Your grade is " + grade)
26      }
27  </script>
28
29  </body>
30  </html>

```

(a) Give the identifier of **two** variables used in the JavaScript code.

1

2

[2]

MARK SCHEME

6(a)	1 mark each mark grade	2
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9608_s18_qp_12_sc_6.b

QUESTION

(b) Give the line number where the JavaScript code produces an output.

.....[1]

MARK SCHEME

6(b)	25	1
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9608_s18_qp_12_sc_6.c

QUESTION

(c) Describe the purpose of the statement on line 11.

.....
.....[2]

MARK SCHEME

6(c)	1 mark per bullet Takes the value entered in the <u>text box / input field</u> 'Mark' Stores (it) in the variable <u>mark</u> // Assigns (it) to the variable <u>mark</u>	2
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9608_s18_qp_12_sc_6.d.i

QUESTION

(d) (i) State whether this JavaScript code will be run client-side or server-side.

.....[1]

MARK SCHEME

6(d)(i)	Client-side	1
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9608_s18_qp_12_sc_6.d.ii

QUESTION

- (ii) Explain the difference between **client-side** scripting and **server-side** scripting.

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.....

.....[3]

MARK SCHEME

6(d)(ii)	1 mark per bullet to max 3 Client-side (script) is run on the computer making the request ...when the (web page) data is received by the computer Server-side (script) is run on the <u>web</u> server The <u>results</u> are sent to the computer that made the request	3
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9608_s18_qp_11_sc_5.a

QUESTION

- 5** A college has a client-server network.

- (a) The college has a file server and other servers.

State the purpose of **two** other servers in the college network.

Server 1

Server 2[2]

MARK SCHEME

5(a)	1 mark per server e.g. E-mail Print Web	2
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QUESTION

- (c) Students use the college's learning resource website. Several of the web pages include PHP script.

Describe the sequence of events when a student requests a web page with embedded server-side code.

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.....[4]

MARK SCHEME

5(c)	1 mark per bullet to max 4 The browser requests the web page The web server accesses the page The web server processes / executes the code The web server produces the HTML for the web page / generates the web page The web server returns the web page to the client The client browser displays this web page	4
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QUESTION

- 7 The design of a web-based application can require the use of client-side scripting.

- (a) Describe what is meant by **client-side scripting**.

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.....[2]

MARK SCHEME

7(a)	Two from: The user's web browser is the client software The requested web page has program code / script embedded <u>within it</u> This code is interpreted by the web browser	1 1 1	2
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9608_s17_qp_13_sc_7.b

QUESTION

- (b) A user requests a web page by keying the Uniform Resource Locator (URL) into the address bar of their web browser.

The requested page contains a client-side script.

Describe the sequence of steps leading to the display of the web page on the computer screen.

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.....[4]

MARK SCHEME

7(b)	Four from: The browser parses the URL to obtain the Domain Name The browser software passes the Domain Name to the nearest Domain Name Server (DNS) The DNS stores a list of Domain Names and matching IP addresses The DNS Name Resolver looks for the Domain Name in its database If found the corresponding IP address is returned to the originator If not found the request is forwarded to another higher level DNS The original DNS adds the returned IP address to its cache The original DNS returns the IP address to the originator The browser uses the IP address to request the required web page from the <u>web server</u> The web server retrieves the page and delivers it to the originator The browser software interprets <u>the script</u> and displays the web page	1 1 1 1 1 1 1 1 1 1 1	Max 4
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9608_s17_qp_12_sc_6.a

QUESTION

6 Downloading a file from a website is an example of a client-server application.

(a) Describe what is meant by the term **client-server** for this application.

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.....

.....

.....[2]

MARK SCHEME

6(a)	Two from: The <u>file</u> is made available from a web/email/FTP server The user's <u>browser</u> is the client software The client (software browser) <u>requests</u> the <u>file</u> from the server The desired <u>file</u> is returned to the client computer	1 1 1 1	Max 2
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9608_s17_qp_11_sc_7.a

QUESTION

7 The design of a web-based application can require the use of client-side scripting.

(a) Describe what is meant by **client-side scripting**.

.....

.....

.....

.....[2]

MARK SCHEME

7(a)	Two from: The user's web browser is the client software The requested web page has program code / script embedded <u>within it</u> This code is interpreted by the web browser	1 1 1	2
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The requested page contains a client-side script.

Describe the sequence of steps leading to the display of the web page on the computer screen.

[4]

MARK SCHEME

7(b)	Four from:	Max 4
	The browser parses the URL to obtain the Domain Name	1
	The browser software passes the Domain Name to the nearest Domain Name Server (DNS)	1
	The DNS stores a list of Domain Names and matching IP addresses	1
	The DNS Name Resolver looks for the Domain Name in its database	1
	If found the corresponding IP address is returned to the originator	1
	If not found the request is forwarded to another higher level DNS	1
	The original DNS adds the returned IP address to its cache	1
	The original DNS returns the IP address to the originator	1
	The browser uses the IP address to request the required web page from the <u>web server</u>	1
	The web server retrieves the page and delivers it to the originator	1
	The browser software interprets <u>the script</u> and displays the web page	1

QUESTION

7 (a) (i) Describe what is meant by a client-server model of networked computers.

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.....

.....

.....[2]

MARK SCHEME

- 7 (a) (i) – at least one computer used to “serve” ...
- ... other computers are referred to as “clients”
- server provides services / applications etc. ...
- ... which may be requested by clients

[2]

QUESTION

(ii) Give **two** benefits of using the client-server model.

1

.....

2

.....

[2]

MARK SCHEME

(ii) any **two** from:

- files and resources are centralised
- creation of security / manage security
- user needs user name and password to access network
- centralised back-up
- intranet capability
- Internet monitoring
- Clients can be less powerful machines, therefore less expensive to buy
- Saving resources on server reduces the burden on the client

[2]

QUESTION

- (c) A web page offers a link for users to request another web page. The requested web page contains HTML code and JavaScript code.

Put each statement in the correct sequence by writing the numbers 1 to 5 in the right-hand column.

Statement	Sequence number
The requested web page is displayed on the client computer	
The user clicks on the hyperlink and the web page is requested from the web server	
The requested web page content is transmitted to the client computer	
The client computer processes the JavaScript code using the web browser software	
The web server locates the requested web page	

[5]

MARK SCHEME

(c)

Statement	Sequence number
The requested web page is displayed on the client computer	5
The user clicks on the hyperlink and the web page is requested from the web server	1
The requested web page content is transmitted to the client computer	3
The client computer processes the JavaScript code using the web browser software	4
The web server locates the requested web page	2

[5]

QUESTION

7 (a) (i) Describe what is meant by a client-server model of networked computers.

.....

.....

.....

.....[2]

MARK SCHEME

- 7 (a) (i) – at least one computer used to “serve” ...
- ... other computers are referred to as “clients”
- server provides services / applications etc. ...
- ... which may be requested by clients

[2]

QUESTION

(ii) Give **two** benefits of using the client-server model.

1

.....

2

.....

[2]

MARK SCHEME

(ii) any **two** from:

- files and resources are centralised
- creation of security / manage security
- user needs user name and password to access network
- centralised back-up
- intranet capability
- Internet monitoring
- clients can be less powerful machines, therefore less expensive to buy
- saving resources on server reduces the burden on the client

[2]

QUESTION

- (c) A web page offers a link for users to request another web page. The requested web page contains HTML code and JavaScript code.

Put each statement in the correct sequence by writing the numbers 1 to 5 in the right-hand column.

Statement	Sequence number
The requested web page is displayed on the client computer	
The user clicks on the hyperlink and the web page is requested from the web server	
The requested web page content is transmitted to the client computer	
The client computer processes the JavaScript code using the web browser software	
The web server locates the requested web page	

[5]

MARK SCHEME

(c)

Statement	Sequence number
The requested web page is displayed on the client computer	5
The user clicks on the hyperlink and the web page is requested from the web server	1
The requested web page content is transmitted to the client computer	3
The client computer processes the JavaScript code using the web browser software	4
The web server locates the requested web page	2

[5]

9608_s15_qp_12_sc_5.c.iii

QUESTION

- (iii) A device or software that provides a specific function for computers using a network. The most common examples handle printing, file storage and the delivery of web pages.

.....[1]

MARK SCHEME

(iii) server

[1]

9608_s15_qp_11_sc_5.c.iii

QUESTION

- (iii) A device or software that provides a specific function for computers using a network. The most common examples handle printing, file storage and the delivery of web pages.

.....[1]

MARK SCHEME

(iii) server

[1]

Show understanding of thin-client and thick-client and the differences between them

Show understanding of the bus, star, mesh and hybrid topologies

Show understanding of cloud computing

Show understanding of the differences between and implications of the use of wireless and wired networks

9608_s21_qp_12_sc_9.a

QUESTION

9 Networks can be either wired using cables or wireless using radio waves.

(a) Describe **one** benefit of using a wireless network compared to using a wired network.

.....

.....

.....

..... [2]

MARK SCHEME

9(a)	1 mark for identification, 1 mark for description (max 2) Benefits: • Less hard-wiring/hardware is required • Users and computers can be mobile • It is much more straightforward to connect other devices Descriptions: • Reduced cost of setting up the network • The network can be accessed from anywhere within range of an access point • No need to physically connect each device	2
------	--	----------

QUESTION

(b) Describe **two** drawbacks of using a wireless network compared to using a wired network.

Drawback 1

.....

.....

.....

Drawback 2

.....

.....

.....

[4]

MARK SCHEME

9(b)	1 mark for identification, 1 mark for matching description (max 2 for each) • Transmissions may be less secure • ...because data packets can be intercepted // easier connection by unauthorised user • Bandwidth may be limited // As more devices connect the bandwidth can be reduced • ...so access may be slow • It is subject to interference from other signals or obstacles • ...which can hinder transmission or corrupt data • Limited range // greater attenuation • ...so there is a need for repeaters // users can easily move out of range • Higher latency • ...so transmission will be slower	4
------	---	----------

QUESTION

- 1 Ana owns a small company with four employees. The office has a network containing several computers that run on a client-server model. There is one server that connects to the Internet using a router.

(a) Networks transmit data using various types of connection shown in the following table.

Complete the table.

Type of connection	Description
Fibre-optic	
.....	A communication device in Earth's orbit that receives and transmits data
Radio waves	
.....	Carries data as electrical signals and can consist of a twisted pair

[4]

MARK SCHEME

1(a)	1 mark for each correctly completed media or description	4										
	<table><tr><th>Type of connection</th><th>Description</th></tr><tr><td>Fibre-optic</td><td>Transmits data as light // Uses (a bundle of) glass/plastic threads to transmit data</td></tr><tr><td>Satellite</td><td>A communication device in Earth's orbit that receives and transmits data</td></tr><tr><td>Radio Waves</td><td>Carries data wirelessly, often known as Wi-Fi // Carries data in the form of electromagnetic waves</td></tr><tr><td>Copper cable</td><td>Carries data as electrical signals and can consist of a twisted pair</td></tr></table>	Type of connection	Description	Fibre-optic	Transmits data as light // Uses (a bundle of) glass/plastic threads to transmit data	Satellite	A communication device in Earth's orbit that receives and transmits data	Radio Waves	Carries data wirelessly, often known as Wi-Fi // Carries data in the form of electromagnetic waves	Copper cable	Carries data as electrical signals and can consist of a twisted pair	
Type of connection	Description											
Fibre-optic	Transmits data as light // Uses (a bundle of) glass/plastic threads to transmit data											
Satellite	A communication device in Earth's orbit that receives and transmits data											
Radio Waves	Carries data wirelessly, often known as Wi-Fi // Carries data in the form of electromagnetic waves											
Copper cable	Carries data as electrical signals and can consist of a twisted pair											

9608_s20_qp_12_sc_7.b

QUESTION

(b) The company wants to upgrade their internet connection to fibre-optic cables.

Give **one** benefit and **one** drawback to the company of upgrading to fibre-optic cables.

Benefit

.....

.....

Drawback

.....

.....

[2]

MARK SCHEME

7(b)	1 mark per bullet point, max 1 for benefit, max 1 for drawback	2
	Benefit: (Consistently) faster data transmission - More stable connection Drawback: - High initial cost as new hardware will be needed - Expertise required to complete connections	

QUESTION

(c) The bank is upgrading its local area network (LAN) copper cables to fibre-optic cables.

(i) State **two** benefits to the bank of upgrading to fibre-optic cable from copper cable.

- 1
-
- 2
-

[2]

MARK SCHEME

4(c)(i)	1 mark per bullet point to max 2 Less interference in the signal The signal does not degrade as quickly // Needs less signal boosting More secure // more difficult to hack Greater bandwidth // <u>Faster</u> transmission speeds possible	2
---------	--	----------

QUESTION

(ii) State **two** drawbacks of upgrading to fibre-optic cables.

- 1
-
- 2
-

[2]

MARK SCHEME

4(c)(ii)	1 mark per bullet point to max 2 Initial installation cost is higher // Cable / hardware is more expensive to buy per metre Specialists / trained personnel are needed to install / maintain Difficult to terminate // The electronics at both ends are more complex Fibres can break <u>when bent</u> Only transmits data in one direction // Cannot transmit power, only data	2
----------	---	----------

QUESTION

- 7 The network manager of a Local Area Network (LAN) has replaced the Ethernet cables with a wireless network.

(a) Give **three** benefits of a wireless network compared to a wired network.

- 1
-
- 2
-
- 3
-
- [3]

MARK SCHEME

7(a)	1 mark per benefit to max 3 • Devices can be more mobile as they do not have to be connected to cable • Easier to set up // no cables need to be installed • Add additional devices is easier • Many different types of device can be connected at the same time	3
------	--	----------

QUESTION

(b) Give **one** drawback of a wireless network compared to a wired network.

-
-[1]

MARK SCHEME

7(b)	1 mark for a drawback from the following: • <u>Easier</u> to hack • Interference • Signal degrades quickly	1
------	---	----------

QUESTION

(c) A telecommunications operator has installed fibre-optic cables in Gopal's neighbourhood.

(i) Give **three** benefits of fibre-optic cable over copper cable.

- 1
-
- 2
-
- 3
-
- [3]

MARK SCHEME

2(c)(i)	1 mark per bullet point to max 3 • Less interference in signal • Signal does not degrade as fast // Needs less signal boosting • More difficult to hack // more secure • Greater bandwidth // <u>Faster</u> transmission speeds possible	3
---------	---	----------

QUESTION

(ii) Give **two** drawbacks of fibre-optic cable over copper cable.

- 1
-
- 2
-
- [2]

MARK SCHEME

2(c)(ii)	1 mark per bullet point to max 2 • (Initial) installation cost is higher // Cable / hardware is more expensive to buy (per metre) • Specialists / trained personnel needed to install / maintain • Difficult to terminate // Electronics at both ends are more complex • Fibre-optic cables can break <u>when bent</u> • Only transmits data in one direction • If a fibre-optic cable connection fails, many more services can be affected	2
----------	---	----------

9608_w18_qp_11_sc_2.c

QUESTION

(c) One type of transmission media is copper cable.

Give **two** additional types of transmission media.

1

2 [2]

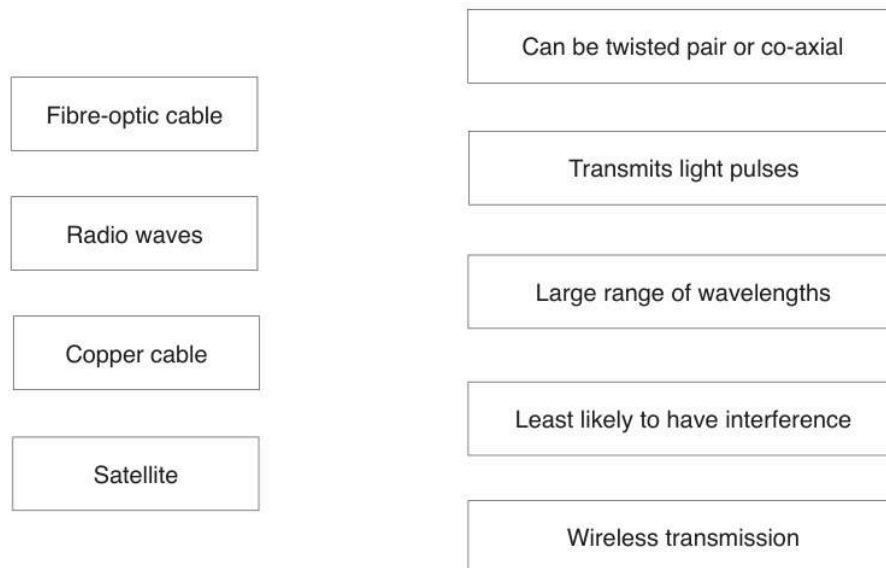
MARK SCHEME

2(c)	1 mark per example to max 2 Fibre-optic Wi-Fi / Radio waves Microwave Infrared	2
------	--	----------

QUESTION

- 1 Four communication media and five features are shown.

Draw one or more lines from each communication media to the appropriate feature(s).

Communication media**Feature**

[6]

MARK SCHEME

1	1 mark for a correct line from each communication media, max 6	6
---	---	---

QUESTION

- (b) Three methods of connecting devices include fibre-optic cables, copper cables and radio waves. The table below gives descriptions relating to these connection methods.

Tick (✓) one box on each row to show the method that best fits each description.

Description	Fibre-optic cable	Copper cable	Radio waves
Wireless medium			
Twisted-pair is an example			
Uses light waves			
WiFi			
Fastest transmission medium			

[5]

MARK SCHEME

(b)

[5]

Description	Fibre-Optic cables	Copper cables	Radio waves
'Wireless' media			✓
Twisted-pair is an example		✓	
Uses light waves	✓		
WiFi			✓
Fastest transmission media	✓		

QUESTION

- 6 A company operates a chemical plant, which has a number of processes. Local computers monitor these processes and collect data.

The computers transfer these data to a central computer 50km away. A telecommunications company (telco) provides cables.

Engineers at the telco had to decide which type of cable to use. They considered the use of either copper cable or fibre optic cable.

State **two** benefits of each type of cable. Each benefit must be clearly different.

Benefits of copper cable

1

2

Benefits of fibre optic cable

1

2

[4]

MARK SCHEME

- 6 any **four** points from (maximum 3 marks per type of cable):

- fibre optic cables have greater bandwidth
- fibre optic cables need less signal boosting // can transmit over longer distances
- fibre optic cables have greater security (more difficult to “tap” into)
- fibre optic cables are immune to electromagnetic and other effects
- fibre optic cabling is lighter in weight (easier to install)
- fibre optic cables consume less power
- copper cabling is less expensive to install
- copper cable is easier to install because it is more flexible
- it is easier to make terminations using copper cabling
- the expertise in use of copper cabling is more extensive
- has been around for years ... so very little is “unknown” about installations using this type of cabling

[4]

QUESTION

- 6** A company operates a chemical plant, which has a number of processes. Local computers monitor these processes and collect data.

The computers transfer these data to a central computer 50km away. A telecommunications company (telco) provides cables.

Engineers at the telco had to decide which type of cable to use. They considered the use of either copper cable or fibre optic cable.

State **two** benefits of each type of cable. Each benefit must be clearly different.

Benefits of copper cable

1

.....

2

.....

Benefits of fibre optic cable

1

.....

2

.....

[4]

MARK SCHEME

- 6** any **four** points from (maximum 3 marks per type of cable):

- fibre optic cables have greater bandwidth
- fibre optic cables need less signal boosting // can transmit over longer distances
- fibre optic cables have greater security (more difficult to “tap” into)
- fibre optic cables are immune to electromagnetic and other effects
- fibre optic cabling is lighter in weight (easier to install)
- fibre optic cables consume less power
- copper cabling is less expensive to install
- copper cable is easier to install because it is more flexible
- it is easier to make terminations using copper cabling
- the expertise in use of copper cabling is more extensive
- has been around for years ... so very little is “unknown” about installations using this type of cabling

[4]

Describe the hardware that is used to support a LAN

9608_s19_qp_12_sc_1.d

QUESTION

(d) The company has an email server.

Identify **three** other types of server.

1

2

3

[3]

MARK SCHEME

1(d)	1 mark per bullet point to max 3 for any valid answer For example: • File server • Print server • Proxy server • Web server • Application server	3
------	---	---

9608_s18_qp_11_sc_5.a

QUESTION

5 A college has a client-server network.

(a) The college has a file server and other servers.

State the purpose of **two** other servers in the college network.

Server 1

Server 2

[2]

MARK SCHEME

5(a)	1 mark per server e.g. E-mail Print Web	2
------	--	----------

9608_s15_qp_12_sc_5.c.iii

QUESTION

- (iii) A device or software that provides a specific function for computers using a network. The most common examples handle printing, file storage and the delivery of web pages.

.....[1]

MARK SCHEME

(iii) server	[1]
--------------	------------

9608_s15_qp_11_sc_5.c.iii

QUESTION

- (iii) A device or software that provides a specific function for computers using a network. The most common examples handle printing, file storage and the delivery of web pages.

.....[1]

MARK SCHEME

(iii) server	[1]
--------------	------------

Describe the role and function of a router in a network

9608_s20_qp_12_sc_7.a

QUESTION

- 7 A company allows customers to stream music from its servers over the Internet.
The company’s internet connection is currently provided through copper cables.
- (a) Identify **two** pieces of hardware, other than the cables, that enable the servers to connect to the Internet. Describe the purpose of each device.
- Device 1
- Purpose
-
-
- Device 2
- Purpose
-
-
- [4]

MARK SCHEME

7(a)	1 mark for device, 1 mark for matching purpose. Max 2 devices. • Router • To connect (devices) / the servers to the Internet // to transmit data between the servers and the Internet // to forward data towards its destination • Gateway • To connect a server that uses a different protocol to the Internet // to join two different types of network • Modem • To connect (the servers) to the Internet over a telephone line • Network interface card // NIC • To enable the servers to connect to the (company) network	4
------	--	----------

QUESTION

- (c) When Bart is at work, he connects his work laptop to his employer's Local Area Network (LAN). The LAN has both a router and a gateway.

Give **two** similarities and **one** difference between a router and a gateway.

Similarity 1

.....

Similarity 2

.....

Difference

.....

.....

[3]

MARK SCHEME

8(c)	1 mark for each similarity, max 2 - Both devices regulate network traffic between two networks // connect two networks - Both receive packets from a network and both forward packets onto a network 1 mark for a difference - A Router connects two networks using the same protocol, a Gateway can connect two networks using different protocols	3
------	---	----------

QUESTION

(c) A network can use routers and gateways.

Explain the role of routers **and** gateways in a network.

.....

.....

.....

.....

.....

.....

.....

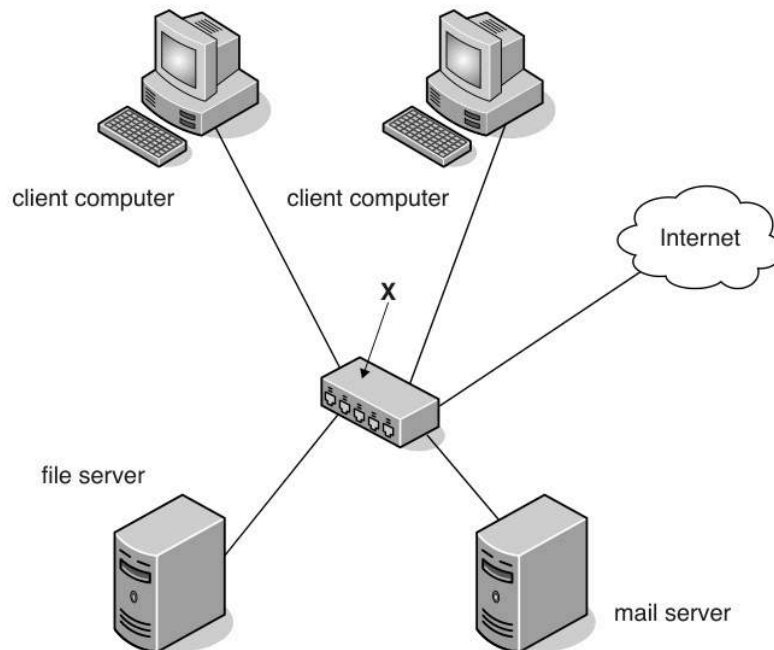
..... [4]

MARK SCHEME

1(c)	1 mark per bullet point to max 4, max 3 for router, max 3 for gateway Only award the repeated bullet points (1 to 5 in each section) once Router: • Connects two (or more) networks • Can connect a network to a WAN // acts as the single access point for... • Receives packets and forwards towards the destination • ...using the IP address of the destination • Assigns private IP addresses • Operates between <u>similar</u> networks // networks using the <u>same protocol</u> • Can be used to segment a network Gateway: • Connect two (or more) networks • Can connect a network to a WAN // acts as the single access point for... • Receives packets and send packets towards the destination • ...using the IP address of the destination • Assigns private IP addresses • Connects two <u>dissimilar</u> networks // networks that use <u>different protocols</u>	4
------	--	----------

QUESTION

(b) The diagram shows a computer network with connection to the Internet.



Name the hardware device labelled X.

.....[1]

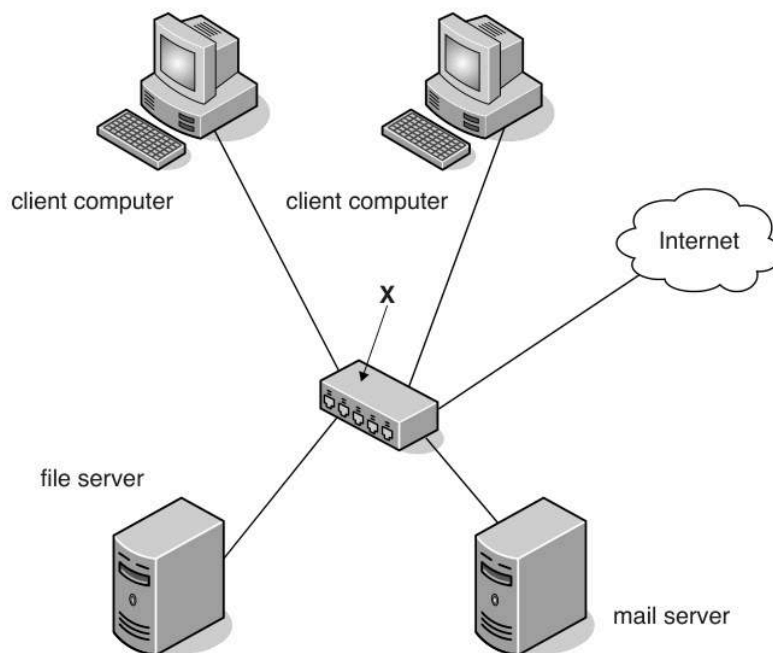
MARK SCHEME

(b) router

[1]

QUESTION

(b) The diagram shows a computer network with connection to the Internet.



Name the hardware device labelled X.

.....[1]

MARK SCHEME

(b) router

[1]

QUESTION

(c) Name the hardware device that is being described:

- (i) A device that transfers data from one network to another in an intelligent way. It has the task of forwarding data packets to their destination by the most efficient route.

.....[1]

MARK SCHEME

(c) (i) router

[1]

QUESTION

(c) Name the hardware device that is being described:

- (i) A device that transfers data from one network to another in an intelligent way. It has the task of forwarding data packets to their destination by the most efficient route.

.....[1]

MARK SCHEME

(c) (i) router

[1]

Show understanding of Ethernet and how collisions are detected and avoided

Show understanding of bit streaming

9608_w19_qp_12_sc_6.e.i

QUESTION

(e) Dominic sends his videos to his colleagues over the Internet using bit streaming.

(i) Describe how the video is sent using bit streaming.

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

6(e)(i)	1 mark per bullet to max 3 The data is compressed before transmitting The video is transmitted continuously as a series of bits The video is hosted on a media server On download, the server sends the data to a buffer on the client computers // The buffer stores the data from the server The recipient / user's software receives bit stream from the buffer	4
---------	---	----------

9608_w19_qp_12_sc_6.e.ii

QUESTION

(ii) Circle either Real-time or on-demand to identify whether the video will be sent using real-time or on-demand bit streaming. Justify your choice.

Real-time / on-demand

Justification

.....

.....

..... [2]

MARK SCHEME

6(e)(ii)	1 mark for: On-demand 1 mark for justification from: The video does not need to be broadcast live // the video is already recorded Dominic's colleagues will watch the video at a later date // at their convenience	2
----------	--	----------

9608_s19_qp_11_sc_1.c

QUESTION

(c) People use the Internet to stream media. Complete the following statements by filling in the names of the missing methods of bit streaming. bit streaming is used when watching a live stream of events that are currently taking place. The event is captured live with a video camera connected to a computer, and it cannot be paused or rewind. bit streaming is used when watching an event that has taken place in the past. Existing media are encoded to bit streaming format and uploaded to a server. It can be paused and rewind. [2]

MARK SCHEME

1(c)	1 mark for each correct term Real-time On-demand	2
------	---	----------

QUESTION

- 6** A user watches a video available on a website. The website uses on-demand bit streaming.

Describe how it is possible to watch the video without it continually pausing.

.....

.....

.....

.....

.....

.....

.....

.....[4]

MARK SCHEME

- 6** Any **four** from:

- User needs high-speed broadband (connection)
 - Data is streamed to a buffer (in the computer)
 - Buffering stops video pausing as bits streamed
 - As buffer is emptied, it fills up again so that viewing is continuous
 - Actual playback is (a few seconds) behind the time the data is received by computer
- [4]

QUESTION

- 6 A user watches a video available on a website. The website uses on-demand bit streaming.

Describe how it is possible to watch the video without it continually pausing.

.....

.....

.....

.....

.....

.....

.....

.....[4]

MARK SCHEME

- 6 Any **four** from:

- User needs high-speed broadband (connection)
- Data is streamed to a buffer (in the computer)
- Buffering stops video pausing as bits streamed
- As buffer is emptied, it fills up again so that viewing is continuous
- Actual playback is (a few seconds) behind the time the data is received by computer [4]

QUESTION

- (c) Bit streaming is used for both real-time and on-demand services.

Describe **one** difference between real-time and on-demand bit streaming.

.....

.....

.....

.....[2]

MARK SCHEME

(c) One pair from:

[2]

- Real-time - a live stream of an event that is currently taking place
- On-demand - streaming of an event/programme that has taken place in the past
- Real time – the event is captured live with a video camera connected to a computer
- On-demand – Existing media are encoded to bit streaming format and uploaded to a server
- Real-time – cannot be paused / rewind etc
- On-demand – can be paused / re-wound / fast forwarded etc

9608_w15_qp_13_sc_1.a

QUESTION

1 (a) Explain the term bit streaming.

.....

.....

.....

.....[2]

MARK SCHEME

1 (a) any two from:

- sequence of digital signals / bits
- over a communication path / Internet
- transfer of data at high speed
- requires fast broadband connection
- requires some form of buffering
- bits arrive in the same order as sent

[2]

QUESTION

(b) A person watches a film streamed from a website on a tablet computer.

(i) Give **two** benefits of using bit streaming for this purpose.

1

.....

2

.....

[2]

MARK SCHEME

(b) (i) any **two** from:

- no need to wait for a whole file to be downloaded
- no need to store large files on user's computer
- allows on demand playback
- no specialist software is required for playback in browser

[2]

QUESTION

(ii) State **two** potential problems of using bit streaming for this purpose.

1

.....

2

.....

[2]

MARK SCHEME

(ii) any **two** from:

- video stops / hangs if very slow Internet / broadband speed low
- video stops / hangs if inadequate buffering capacity
- loss of Internet means can't access films / files
- may require specific software to run the files / films
- viruses can be downloaded from the websites

[2]

QUESTION

(c) Explain the terms on-demand bit streaming and real-time bit streaming.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

MARK SCHEME

(c) 2 marks for on-demand and 2 marks for real-time

on-demand

- digital video tape, analogue video tape, or digital files are converted to bit streaming – format for broadcasting on the net; this is known as encoding these encoded streaming video files are then uploaded to a dedicated server
- a link for the encoded video is placed on a web site
- a user clicks on the link to download the encoded streaming video; the streamed video is then broadcast to the user as and when they require it
- can be paused / can go back and re-watch / fast-forward, etc.

real-time

- an event is captured live with a video camera
- the video camera is connected to a computer
- the video signal is converted to streaming media files (encoded) on the computer
- the encoded feed is then uploaded from the computer to a dedicated streaming server via cable, DSL, or a high-speed internet connection
- the server then sends the live images it to all users requesting it as real-time video streaming
- cannot be paused etc.

[4]

9608_w15_qp_11_sc_1.a

QUESTION

- 1 (a) Explain the term bit streaming.

.....

.....

.....

.....[2]

MARK SCHEME

- 1 (a) any **two** from:

- sequence of digital signals / bits
- over a communication path / Internet
- transfer of data at high speed
- requires fast broadband connection
- requires some form of buffering
- bits arrive in the same order as sent

[2]

9608_w15_qp_11_sc_1.b.i

QUESTION

- (b) A person watches a film streamed from a website on a tablet computer.

- (i) Give **two** benefits of using bit streaming for this purpose.

1

.....

2

.....[2]

MARK SCHEME

- (b) (i) any **two** from:

- no need to wait for a whole file to be downloaded
- no need to store large files on user's computer
- allows on demand playback
- no specialist software is required for playback in browser

[2]

MARK SCHEME

(c) 2 marks for on-demand and 2 marks for real-time

on-demand

- digital video tape, analogue video tape, or digital files are converted to bit streaming – format for broadcasting on the net; this is known as encoding, these encoded streaming video files are then uploaded to a dedicated server
- a link for the encoded video is placed on a web site
- a user clicks on the link to download the encoded streaming video; the streamed video is then broadcast to the user as and when they require it
- can be paused / can go back and re-watch / fast-forward, etc.

real-time

- an event is captured live with a video camera
- the video camera is connected to a computer
- the video signal is converted to streaming media files (encoded) on the computer
- the encoded feed is then uploaded from the computer to a dedicated streaming server via cable, DSL, or a high-speed internet connection
- the server then sends the live images it to all users requesting it as real-time video streaming
- cannot be paused etc.

[4]

Show understanding of the differences between the World Wide Web (WWW) and the internet

9608_s18_qp_11_sc_5.b

QUESTION

(b) The students use the network to access the Internet.

One student stated, '*The Internet and the World Wide Web are the same thing*'.

Tick (✓) **one** box to indicate whether this statement is true or false.

True	False
☐	☐

Justify your choice.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[5]

MARK SCHEME

5(b)	1 mark for the indicating the statement is false: <table><tr><td>True</td><td>False</td></tr><tr><td></td><td>✓</td></tr></table> 1 mark per bullet for justification to max 4. - Internet is the infrastructure / global collection of networks - World Wide Web is the (multimedia web) pages / content - The World Wide Web is accessed over the Internet - Webpages are written in HTML - HTTP protocol used to transfer web pages - Internet uses IP protocol	True	False		✓	5
True	False					
	✓					

9608_s18_qp_11_sc_5.c

QUESTION

(c) Students use the college's learning resource website. Several of the web pages include PHP script.

Describe the sequence of events when a student requests a web page with embedded server-side code.

.....

.....

.....

.....

.....

.....

.....

.....[4]

MARK SCHEME

5(c)	1 mark per bullet to max 4 - The <u>browser</u> requests the web page - The web server accesses the page - The web server processes / executes the code - The web server produces the HTML for the web page / generates the web page - The web server returns the web page to the client - The client browser displays this web page	4
------	--	---

QUESTION

- 6 (a) Explain the difference between the World Wide Web (WWW) and the Internet.

.....

.....

.....

.....[2]

MARK SCHEME

- 6 (a) Two from:

[2]

- WWW is a collection of interlinked, hypertext documents/webpages/multimedia resources (accessed via the Internet) //WWW is content from web servers organised as web pages
- Internet is the global connection of interconnected computer networks
- The Internet uses TCP/IP protocol / WWW uses http protocols to transmit data

QUESTION

- (c) A web page offers a link for users to request another web page. The requested web page contains HTML code and JavaScript code.

Put each statement in the correct sequence by writing the numbers 1 to 5 in the right-hand column.

Statement	Sequence number
The requested web page is displayed on the client computer	
The user clicks on the hyperlink and the web page is requested from the web server	
The requested web page content is transmitted to the client computer	
The client computer processes the JavaScript code using the web browser software	
The web server locates the requested web page	

[5]

MARK SCHEME

(c)

Statement	Sequence number
The requested web page is displayed on the client computer	5
The user clicks on the hyperlink and the web page is requested from the web server	1
The requested web page content is transmitted to the client computer	3
The client computer processes the JavaScript code using the web browser software	4
The web server locates the requested web page	2

[5]

9608_s15_qp_12_sc_5.b

QUESTION

(b) Distinguish between the Internet and the World Wide Web (WWW).

.....
.....
.....
.....
.....
.....[3]

MARK SCHEME

(b) maximum of two marks for Internet references and **maximum of two marks** for world wide web references

Internet

- massive network of networks/interconnected network of computer devices
- Internet stands for Interconnected Networks
- uses TCP/IP protocol

World Wide Web (www)

- is a collection of (multimedia) web pages/documents
- ...stored on websites
- http/protocols used to transmit data
- web pages are written in HTML
- URLs specify the location of the web pages
- web documents are accessed using browsers

[3]

9608_s15_qp_11_sc_5.b

QUESTION

(b) Distinguish between the Internet and the World Wide Web (WWW).

.....

.....

.....

.....

.....

.....[3]

MARK SCHEME

(b) maximum of two marks for Internet references and **maximum of two marks** for world wide web references

Internet

- massive network of networks/interconnected network of computer devices
- Internet stands for Interconnected Networks
- uses TCP/IP protocol

World Wide Web (www)

- is a collection of (multimedia) web pages/documents
- ...stored on websites
- http/protocols used to transmit data
- web pages are written in HTML
- URLs specify the location of the web pages
- web documents are accessed using browsers

[3]

Describe the hardware that is used to support the internet

9608_s20_qp_12_sc_7.a

QUESTION

- 7

A company allows customers to stream music from its servers over the Internet. The company's internet connection is currently provided through copper cables.

(a) Identify **two** pieces of hardware, other than the cables, that enable the servers to connect to the Internet. Describe the purpose of each device.

Device 1

Purpose

.....

.....

Device 2

Purpose

.....

.....

[4]

MARK SCHEME

7(a)	1 mark for device, 1 mark for matching purpose. Max 2 devices. - Router - To connect (devices) / the servers to the Internet // to transmit data between the servers and the Internet // to forward data towards its destination - Gateway - To connect a server that uses a different protocol to the Internet // to join two different types of network - Modem - To connect (the servers) to the Internet over a telephone line - Network interface card // NIC - To enable the servers to connect to the (company) network	4
------	---	---

9608_s19_qp_12_sc_1.b.i

QUESTION

(b) A company has computers in two separate buildings that communicate using the Internet over a Public Switched Telephone Network (PSTN).

(i) Describe the transmission of data using a PSTN.

.....

.....

.....

..... [2]

MARK SCHEME

1(b)(i)	1 mark per bullet point to max 2 • The PSTN consists of many different types of communication lines • Data is transmitted in both directions <u>at the same time</u> // (full) <u>duplex</u> data transmission • The communication passes through different switching centres	2
---------	---	----------

9608_s19_qp_12_sc_1.b.ii

QUESTION

(ii) The company wants to install a dedicated line between the two buildings.

Identify **one** benefit and **one** drawback of installing a dedicated line between the two buildings.

Benefit

.....

Drawback

.....

[2]

MARK SCHEME

1(b)(ii)	1 mark for benefit, 1 mark for drawback Benefit • (Probably) faster connection / communication / transmission of data • (Usually) more consistent transmission speed • Improved security Drawback • Expensive to <u>set-up / maintain</u> • Disruption to the dedicated line would leave no alternative	2
----------	---	----------

9608_s15_qp_12_sc_5.a

QUESTION

5 (a) Telephone calls can be made by using:

- conventional telephones (using the Public Service Telephone Network (PSTN) system) over a wired network
- a computer, equipped with speakers and microphone, connected to the Internet

Put a tick (✓) in the correct column to match each description to the appropriate communication method.

Description	Conventional telephone using PSTN	Internet-based system
connection only in use whilst sound is being transmitted		
dedicated channel used between two points for the duration of the call		
connection maintained throughout the telephone call		
encoding schemes and compression technology used		
lines remain active even during a power outage		

[5]

MARK SCHEME

5 (a)

Description	Conventional telephone using PSTN	Internet-based system
connection only in use whilst sound is being transmitted		✓
dedicated channel used between two points for the duration of the call	✓	
connection maintained throughout the telephone call	✓	
encoding schemes and compression technology used		✓
lines remain active even during a power outage	✓	

[5]

QUESTION

5 (a) Telephone calls can be made by using:

- conventional telephones (using the Public Service Telephone Network (PSTN) system) over a wired network
- a computer, equipped with speakers and microphone, connected to the Internet

Put a tick (✓) in the correct column to match each description to the appropriate communication method.

Description	Conventional telephone using PSTN	Internet-based system
connection only in use whilst sound is being transmitted		
dedicated channel used between two points for the duration of the call		
connection maintained throughout the telephone call		
encoding schemes and compression technology used		
lines remain active even during a power outage		

[5]

MARK SCHEME

5 (a)

Description	Conventional telephone using PSTN	Internet-based system
connection only in use whilst sound is being transmitted		✓
dedicated channel used between two points for the duration of the call	✓	
connection maintained throughout the telephone call	✓	
encoding schemes and compression technology used		✓
lines remain active even during a power outage	✓	

[5]

Explain the use of IP addresses in the transmission of data over the internet

9608_w21_qp_11_sc_11.b

QUESTION

(b) Computers on the Internet have IP addresses.

Describe the format of an IP address.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

11(b)	One mark per bullet point to max 3 IPv4 • Four groups of (denary or Hexadecimal integers • Numbers between 0 and 255 / 0 and FF • Each stored in 1 byte / 8 bits // the whole is stored in 32 bits / 4 bytes • Separated by full stops • Correct example OR IPv6 • Eight groups of (Hexadecimal) digits • Numbers between 0 and FFFF • Each stored in 2 bytes/16 bits // the whole stored in 128 bits / 16bytes • Separated by colons • The first instance of multiple groups of zero can be replaced by a double colon • correct example	3
-------	--	---

QUESTION

(b) The web page 9608.html is accessed from the URL:

`https://www.cambridgeinternational.org/9608.html`

An employee of the company states, "A Domain Name and an IP address are exactly the same thing".

State whether this statement is true or false **and** justify your choice.

.....

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

6(b)	1 mark per bullet point for justification to max 3 • IP address is (physical) address of the server • A domain name is a memorable form for an IP address • <code>cambridgeinternational.org</code> is the domain name • An example of an IP address is 198.162.2.1 • Each domain name is linked to an IP address • A Domain Name Server/Service (DNS) is used to translate a domain name into its corresponding IP address • Domain name does not change but IP address could change (dynamic) • IP address can be used to address server directly, • The domain name needs translation before server can be assessed	3
------	---	----------

QUESTION

9 (a) Identify **two** differences between a public IP address and a private IP address.

1

.....

2

.....

[2]

MARK SCHEME

9(a)	1 mark per difference - Private IP is only known within the LAN // Public IP is known outside of the LAN/ on Internet - Public is allocated by ISP // Private is allocated by the router - Public addresses are unique throughout the Internet, private addresses are unique only within the LAN - Private IP addresses are more secure than public IP addresses	2
------	--	----------

9608_s20_qp_13_sc_1.c

QUESTION

(c) Each computer in the network has a private IP address.

Give **two** reasons why the computers do **not** have public IP addresses.

1

.....

2

.....

[2]

MARK SCHEME

1(c)	1 mark per bullet point - Improved security because the IP address is not visible outside the network - An internet presence is not required for each employee computer - Only the router needs a public IP address, as only the router needs to be externally visible - Reduces number of (public) IP addresses needed	2
------	---	----------

9608_s19_qp_12_sc_1.a.i

QUESTION

1 Computers on the Internet have IP addresses.

(a) IP addresses can be in either IPv4 or IPv6 format.

(i) Give an example of a valid IPv4 address.

.....

..... [1]

MARK SCHEME

1(a)(i)	1 mark for any valid example e.g. 192.168.0.1	1
---------	---	----------

9608_s19_qp_12_sc_1.a.ii

QUESTION

(ii) State why there is a need for IPv6 addressing.

.....

..... [1]

MARK SCHEME

1(a)(ii)	1 mark for correct answer The number of IP addresses needed will exceed the number available using IPv4.	1
----------	--	----------

9608_s19_qp_12_sc_1.a.iii

QUESTION

(iii) A computer's IPv6 address is:

C100:2235::1000:25AA:AA50

Explain why this IPv6 address would be an invalid IPv4 address.

.....

.....

.....

..... [2]

MARK SCHEME

1(a)(iii)	1 mark per bullet point to max 2 - Too many digits per group - Too many groups of digits - The address is more than 32 bits / 4 bytes - Colons are used as separators	2
-----------	--	----------

QUESTION

1 Devices connected to the Internet have IP (Internet Protocol) addresses.

(a) Three IPv4 addresses are given.

Circle either Valid or Invalid to indicate whether each address is valid or invalid. Explain your decision.

Address 1: 3A.21.2H.1 Valid / Invalid

Explanation

.....

Address 2: 299.53.2.2 Valid / Invalid

Explanation

.....

Address 3: 192.2.1.0 Valid / Invalid

Explanation

.....

[3]

MARK SCHEME

1(a)	1 mark for each correct indication and explanation 3A.21.2H.1 Invalid H is not a valid hexadecimal digit 299.53.2.2 Invalid 299 is not in the correct range 192.2.1.0 Valid Consists of four numbers in the range 0–255 separated by full stops	3
------	---	----------

9608_w18_qp_13_sc_4.b.i

QUESTION

(b) (i) An IPv4 address has been entered as 12.258.3

Give **two** reasons why this IP address is invalid.

1

.....

2

.....

[2]

MARK SCHEME

4(b)(i)	1 mark per bullet point	2
	• <u>258</u> is too large/largest individual numbers is 255 • 4 numbers needed/1 number missing/only 3 groups of numbers given	

9608_w18_qp_13_sc_4.b.ii

QUESTION

(ii) An IPv6 address has been entered as 15EF:5L63::2014:BB::60AA

Give **two** reasons why this IP address is invalid.

1

.....

2

.....

[2]

MARK SCHEME

4(b)(ii)	1 mark per bullet point	2
	• L not a valid hexadecimal number • Only one double colon is allowed	

QUESTION

(c) The table shows four descriptions of IP addresses.

Tick (✓) **one** box in each row to identify whether each description applies to a public or private IP address.

Description	Public	Private
The address can be reached over the Internet.		
The address is more secure.		
The address can only be accessed through the same LAN.		
The address can be duplicated in different networks.		

[4]

MARK SCHEME

4(c)	1 mark per row	4															
<table> <tr> <th>Description</th><th>Public</th><th>Private</th></tr> <tr> <td>The address can be reached over the Internet.</td><td>✓</td><td></td></tr> <tr> <td>The address is more secure.</td><td></td><td>✓</td></tr> <tr> <td>The address can only be accessed through the same LAN.</td><td></td><td>✓</td></tr> <tr> <td>The address can be duplicated in different networks.</td><td></td><td>✓</td></tr> </table>			Description	Public	Private	The address can be reached over the Internet.	✓		The address is more secure.		✓	The address can only be accessed through the same LAN.		✓	The address can be duplicated in different networks.		✓
Description	Public	Private															
The address can be reached over the Internet.	✓																
The address is more secure.		✓															
The address can only be accessed through the same LAN.		✓															
The address can be duplicated in different networks.		✓															

QUESTION

(b) Describe the purpose of an IP address.

.....

.....

.....

..... [2]

MARK SCHEME

2(b)	1 mark per bullet point to max 2 • Gives each device on a network an identifier // IP address used to locate a device on a network • Each address is <u>unique</u> within the network • Allows a device/gateway/node to send data to the correct destination / a specific device/gateway/node	2
------	---	----------

9608_w18_qp_11_sc_2.b.i

QUESTION

(b) (i) The following table shows four IPv6 addresses.

State if each address is valid or invalid.

IP address	Valid or invalid
21E5:69AA:FFFF:1:E100:B691:1285:F56E	
::255.255.255.255	
59FB::1005:CC57:6571	
56FE::2159:5BBC::6594	

[4]

MARK SCHEME

2(b)(i)	1 mark per correct answer		4
	IP Address	Valid or invalid	
	21E5:69AA:FFFF:1:E100:B691:1285:F56E	Valid	
	::255.255.255.255	Valid	
	59FB::1005:CC57:6571	Valid	
56FE::2159:5BBC::6594	Invalid		

QUESTION

(ii) The following table shows four statements about either public or private IP addresses.

Tick (✓) **one** box in each row to indicate whether each statement refers to a public or a private IP address.

Statement	Public	Private
192.168.2.1 is an example of this type of address		
Assigned by the Internet Service Provider (ISP)		
IP address cannot be duplicated in different networks		
Network Address Translation (NAT) is necessary to access the Internet directly		

[4]

MARK SCHEME

2(b)(ii)	1 mark per correct row	4															
<table> <tr> <th>Statement</th><th>Public</th><th>Private</th></tr> <tr> <td>192.168.2.1 is an example of this type of address</td><td></td><td>✓</td></tr> <tr> <td>Assigned by the Internet Service Provider (ISP)</td><td>✓</td><td></td></tr> <tr> <td>IP address cannot be duplicated in different networks</td><td>✓</td><td></td></tr> <tr> <td>Network Address Translation (NAT) is necessary to access the Internet directly</td><td></td><td>✓</td></tr> </table>			Statement	Public	Private	192.168.2.1 is an example of this type of address		✓	Assigned by the Internet Service Provider (ISP)	✓		IP address cannot be duplicated in different networks	✓		Network Address Translation (NAT) is necessary to access the Internet directly		✓
Statement	Public	Private															
192.168.2.1 is an example of this type of address		✓															
Assigned by the Internet Service Provider (ISP)	✓																
IP address cannot be duplicated in different networks	✓																
Network Address Translation (NAT) is necessary to access the Internet directly		✓															

QUESTION

- (d) A device needs an IP address to connect to the Internet. IPv4 is the more common type of IP address.

Describe, using an example, the format of an IPv4 address.

.....

.....

.....

.....

.....

.....[3]

MARK SCHEME

- (d) **Two marks** for description, **one mark** for correct example.

[3]

- Four numbers separated with ‘.’
- Each number is between 0 and 255 / 00 and FF in Hex / stored in one byte.
- 32 bits long
- Correct example

QUESTION

- 7 Access to World Wide Web content uses IP addressing.

- (a) State what IP stands for.

.....[1]

MARK SCHEME

- 7 (a) **Internet Protocol**

[1]

QUESTION

(b) The following table shows four possible IP addresses.

Indicate for each IP address whether it is valid or invalid and give a reason.

Address	Denary / Hexadecimal	Valid or Invalid	Reason
3.2A.6AA.BBBB	Hexadecimal		
2.0.255.1	Denary		
6.0.257.6	Denary		
A.78.F4.J8	Hexadecimal		

[4]

MARK SCHEME

(b)

[4]

Address	Denary / Hexadecimal	Valid or Invalid	Reason
3.2A.6AA.BBBB	Hexadecimal	Invalid	One point from: - This is more than <u>32 bits</u> <u>6AA / BBBB</u> in Hex is bigger than <u>FF / 255</u> in denary <u>6AA / BBBB</u> uses more than 8 bits / a byte - The third / fourth group is bigger than <u>FF / 255</u> in denary - The third / fourth group uses more than 8 bits / a byte
2.0.255.1	Denary	Valid	There are 4 bytes, each 255 or below // All the values are in the range 0 - 255
6.0.257.6	Denary	Invalid	<u>257</u> is above 255 // The third group is above 255
0A.78.F4.J8	Hexadecimal	Invalid	J is not a valid hexadecimal digit // J8 is not a valid Hex number

One mark for each combination of valid or invalid **and** the reason.

9608_s16_qp_12_sc_7.c

QUESTION

(c) Describe **two** differences between public and private IP addresses.

- 1
-
- 2
-[2]

MARK SCHEME

(c) Two points from:

[2]

- Public address can be reached across the Internet.
- Private address can only be reached internally/through the LAN/Intranet // private address cannot be reached across the Internet.
- NAT (Network Address Translation) is necessary for a private IP address to access the Internet directly.
- A private address is more secure than a public address // A public address is less secure than a private address.
- Public addresses are provided by ISP / assigned by InterNIC // Private addresses are assigned by the router (of the network concerned).
- Public addresses are unique (to the Internet) // Private addresses (are unique within their network, but) can be duplicated within other (discrete) networks.
- 10.0.0.1 to 10.255.255.254 and 172.16.0.1 to 172.31.255.254 and 192.168.0.1 to 192.168.255.254 form the private address space // IP addresses from the private address space are never assigned as public.

9608_s16_qp_11_sc_7.a

QUESTION

7 Access to World Wide Web content uses IP addressing.

(a) State what IP stands for.

.....[1]

MARK SCHEME

7 (a) Internet Protocol

[1]

QUESTION

(b) The following table shows four possible IP addresses.

Indicate for each IP address whether it is valid or invalid and give a reason.

Address	Denary / Hexadecimal	Valid or Invalid	Reason
3.2A.6AA.BBBB	Hexadecimal		
2.0.255.1	Denary		
6.0.257.6	Denary		
A.78.F4.J8	Hexadecimal		

[4]

MARK SCHEME

(b)

[4]

Address	Denary / Hexadecimal	Valid or Invalid	Reason
3.2A.6AA.BBBB	Hexadecimal	Invalid	One point from: - This is more than <u>32 bits</u> <u>6AA / BBBB</u> in Hex is bigger than <u>FF / 255</u> in denary <u>6AA / BBBB</u> uses more than 8 bits / a byte - The third / fourth group is bigger than <u>FF / 255</u> in denary - The third / fourth group uses more than 8 bits / a byte
2.0.255.1	Denary	Valid	There are 4 bytes, each 255 or below // All the values are in the range 0 - 255
6.0.257.6	Denary	Invalid	<u>257</u> is above 255 // The third group is above 255
0A.78.F4.J8	Hexadecimal	Invalid	J is not a valid hexadecimal digit // J8 is not a valid Hex number

One mark for each combination of valid or invalid **and** the reason.

9608_s16_qp_11_sc_7.c

QUESTION

(c) Describe **two** differences between public and private IP addresses.

- 1
-
- 2
-[2]

MARK SCHEME

(c) Two points from:

[2]

- Public address can be reached across the Internet.
- Private address can only be reached internally/through the LAN/Intranet // private address cannot be reached across the Internet.
- NAT (Network Address Translation) is necessary for a private IP address to access the Internet directly.
- A private address is more secure than a public address // A public address is less secure than a private address.
- Public addresses are provided by ISP / assigned by InterNIC // Private addresses are assigned by the router (of the network concerned).
- Public addresses are unique (to the Internet) // Private addresses (are unique within their network, but) can be duplicated within other (discrete) networks.
- 10.0.0.1 to 10.255.255.254 and 172.16.0.1 to 172.31.255.254 and 192.168.0.1 to 192.168.255.254 form the private address space // IP addresses from the private address space are never assigned as public.

9608_w15_qp_13_sc_3.a

QUESTION

3 (a) The table shows four statements about IP addresses.

Tick (✓) to show which of the statements are true.

Statement	True (✓)
The IP address consists of any number of digits separated by single dots (.)	
Each number in an IP address can range from 0 to 255	
IP addresses are used to ensure that messages and data reach their correct destinations	
Public IP addresses are considered to be more secure than private IP addresses	

[2]

MARK SCHEME

3 (a)

Statement	True (✓)
The IP address consists of any number of digits separated by single dots (.)	
Each number in an IP address can range from 0 to 255	✓
IP addresses are used to ensure that messages and data reach their correct destinations	✓
Public IP addresses are considered to be more secure than private IP addresses	

accept words TRUE or FALSE in right hand column

1 mark per tick, –1 mark for each wrong tick if more than 2

[2]

9608_w15_qp_11_sc_3.a

QUESTION

3 (a) The table shows four statements about IP addresses.

Tick (✓) to show which of the statements are true.

Statement	True (✓)
The IP address consists of any number of digits separated by single dots (.)	
Each number in an IP address can range from 0 to 255	
IP addresses are used to ensure that messages and data reach their correct destinations	
Public IP addresses are considered to be more secure than private IP addresses	

[2]

MARK SCHEME

3 (a)

Statement	True (✓)
The IP address consists of any number of digits separated by single dots (.)	
Each number in an IP address can range from 0 to 255	✓
IP addresses are used to ensure that messages and data reach their correct destinations	✓
Public IP addresses are considered to be more secure than private IP addresses	

accept words TRUE or FALSE in right hand column

1 mark per tick, –1 mark for each wrong tick if more than 2

[2]

Explain how a Uniform Resource Locator (URL) is used to locate a resource on the World Wide Web (WWW) and the role of the Domain Name Service (DNS)

9608_s21_qp_13_sc_6.b

QUESTION

(b) The web page 9608.html is accessed from the URL:

`https://www.cambridgeinternational.org/9608.html`

An employee of the company states, "A Domain Name and an IP address are exactly the same thing".

State whether this statement is true or false **and** justify your choice.

.....

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

6(b)	1 mark per bullet point for justification to max 3 • IP address is (physical) address of the server • A domain name is a memorable form for an IP address • <code>cambridgeinternational.org</code> is the domain name • An example of an IP address is 198.162.2.1 • Each domain name is linked to an IP address • A Domain Name Server/Service (DNS) is used to translate a domain name into its corresponding IP address • Domain name does not change but IP address could change (dynamic) • IP address can be used to address server directly, • The domain name needs translation before server can be assessed	3
------	---	----------

QUESTION

- (b) A website can be accessed using either the Uniform Resource Locator (URL) or the IP address.

Describe how a URL is converted into its matching IP address.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

1(b)	1 mark per bullet point to max 3 URL is parsed to obtain the Domain name Domain name is sent to the nearest Domain Name Server (DNS) DNS holds a list of Domain names and matching IP addresses DNS name resolver searches its database for the Domain name If DNS does not find the Domain name, the request is forwarded to a higher level DNS If the Domain name is found, the IP address is returned If the Domain name is not found, the request is passed to a higher level server If the Domain name is finally not found, an error message is generated	3
------	--	----------

QUESTION

4 Ava needs to view a website and she knows the Uniform Resource Locator (URL).

(a) Complete the series of steps that take place.

Write the **letter** of the appropriate statement in each space.

A	DNS finds corresponding IP
B	DNS looks up URL in table
C	Ava types the URL into a web browser

- 1
- 2 Web browser sends URL to Domain Name Service (DNS)
- 3
- 4
- 5 DNS returns IP address to web browser

[2]

MARK SCHEME

4(a)	1 mark for 1 letter in correct space 2 marks for all 3 letters in correct places	2
	1 C 2 URL goes to Domain Name Service (DNS) 3 B 4 A 5 DNS returns IP address to client	

QUESTION

- 2 Gopal types the Uniform Resource Locator (URL) of a website into a web browser.
- (a) The following sequence (1 to 5) describes the steps that take place. There are three missing statements.
- 1 Gopal types into the web browser.
 - 2
 - 3 DNS looks up the URL in table
 - 4
 - 5

Three statements **A**, **B** and **C** are used to complete the sequence.

A	DNS finds corresponding IP address
B	Web browser sends URL to Domain Name Service (DNS)
C	DNS returns IP address to web browser

Write one of the letters **A** to **C** in the appropriate rows (2, 4 and 5) to complete the sequence.
[2]

MARK SCHEME

2(a)	1 mark for 1 correct answer, 2 marks for all 3 correct answers	2
	1 Gopal types into the web browser	
	2 B (Web browser sends URL to Domain name Service (DNS))	
	3 DNS looks up URL in a table	
	4 A (DNS finds corresponding IP address)	
	5 C (DNS returns IP address to web browser)	

QUESTION

- 2 Computer **A** needs to access a web page.
- (a) State how Computer **A** could access the web page without using a Domain Name Service (DNS).
-
-[1]

MARK SCHEME

2(a)	Use the IP address instead of the URL	1
------	---------------------------------------	---

9608_s17_qp_13_sc_7.b

QUESTION

- (b) A user requests a web page by keying the Uniform Resource Locator (URL) into the address bar of their web browser.

The requested page contains a client-side script.

Describe the sequence of steps leading to the display of the web page on the computer screen.

[4]

MARK SCHEME

7(b)	Four from:	Max 4
	The browser parses the URL to obtain the Domain Name	1
	The browser software passes the Domain Name to the nearest Domain Name Server (DNS)	1
	The DNS stores a list of Domain Names and matching IP addresses	1
	The DNS Name Resolver looks for the Domain Name in its database	1
	If found the corresponding IP address is returned to the originator	1
	If not found the request is forwarded to another higher level DNS	1
	The original DNS adds the returned IP address to its cache	1
	The original DNS returns the IP address to the originator	1
	The browser uses the IP address to request the required web page from the <u>web server</u>	1
	The web server retrieves the page and delivers it to the originator	1
	The browser software interprets <u>the script</u> and displays the web page	1

QUESTION

- (b) The following sequence of steps (1 to 5) describes what happens when someone uses their personal computer (PC) to request a web page. The web page consists of HTML tags and text content only. Four of the statements from **A**, **B**, **C**, **D**, **E** and **F** are used to complete the sequence.

A	Browser software interprets the script, renders the page and displays.
B	Browser software renders the page and displays.
C	Browser software compiles the script, renders the page and displays.
D	The web server retrieves the page.
E	The Domain Name Service (DNS) uses the domain name from the browser to look up the IP address of the web server.
F	The web server sends the web page content to the browser.

Write one of the letters A to F in the appropriate row to complete the sequence.

1. The user keys in the Uniform Resource Locator (URL) into the browser software.
2.
3.
4.
5.

[4]

MARK SCHEME

6(b)	1. The user keys in the Uniform Resource Locator (URL) into the browser Software. 2. E // The Domain Name Service (DNS) uses the domain name from the browser to look up the IP address of the web server. 3. D // The web server retrieves the page 4. F // Sends the web page content to the browser 5. B // Browser software renders the page and displays	4 1 1 1 1
------	--	---

.....[4]

QUESTION

- (e) A computer user keys in the Uniform Resource Locator (URL) of a web page into a web browser.

Describe how the browser uses the Domain Name Service (DNS) to display the web page.

.....

.....

.....

.....

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.....

.....

.....[4]

MARK SCHEME

(e) Four from:

[4]

- URL is a reference address to a resource on the Internet.
- The URL is passed to the nearest Domain Name Server (by browser software).
- DNS server stores a database / list of URLs and matching IP addresses.
- DNS (Name Resolver) looks for the URL in its database.
- Finds the matching IP address and returns it to the originator.
- Or if it cannot find it, it forwards to another Domain Name Server at a higher level.
- (Original) DNS server adds the returned IP address to its cache.
- (Original) DNS server returns the IP address to the browser.

QUESTION

(b) Consider the URL:

`http://cie.org.uk/computerscience.html`

(i) Give the meaning of the following parts of the URL.

`http`

.....

.....

`cie.org.uk`

.....

.....

`computerscience.html`

.....

.....

[3]

MARK SCHEME

(b) (i) `http` – enables browser to know what protocol is being used to access information in the domain

`cie.org.uk` – `cie.org.uk` is the domain name

`computerscience.html` – actual web page / file being viewed

[3]

QUESTION

(ii) Sometimes the URL contains the characters `%20` and `?`.

Describe the function of these characters.

`%20`

.....

`?`

.....

[2]

MARK SCHEME

(ii) **%20** – because <space> not allowed in a URL, %20 is the coding for a space (32 in denary)

? – separates the URL from all parameters or variables

[2]

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QUESTION

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cie.org.uk

.....

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.....

[3]

MARK SCHEME

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cie.org.uk – cie.org.uk is the domain name

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QUESTION

- (ii) Sometimes the URL contains the characters %20 and ?.

Describe the function of these characters.

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.....

?

.....

[2]

MARK SCHEME

- (ii) %20 – because <space> not allowed in a URL, %20 is the coding for a space (32 in denary)

? – separates the URL from all parameters or variables

[2]

QUESTION

- (c) A web page offers a link for users to request another web page. The requested web page contains HTML code and JavaScript code.

Put each statement in the correct sequence by writing the numbers 1 to 5 in the right-hand column.

Statement	Sequence number
The requested web page is displayed on the client computer	
The user clicks on the hyperlink and the web page is requested from the web server	
The requested web page content is transmitted to the client computer	
The client computer processes the JavaScript code using the web browser software	
The web server locates the requested web page	

[5]

MARK SCHEME

(c)

Statement	Sequence number
The requested web page is displayed on the client computer	5
The user clicks on the hyperlink and the web page is requested from the web server	1
The requested web page content is transmitted to the client computer	3
The client computer processes the JavaScript code using the web browser software	4
The web server locates the requested web page	2

[5]